Can AI Learn Causal Structure?: Evidence from ADIA Lab's Causal Discovery Challenge

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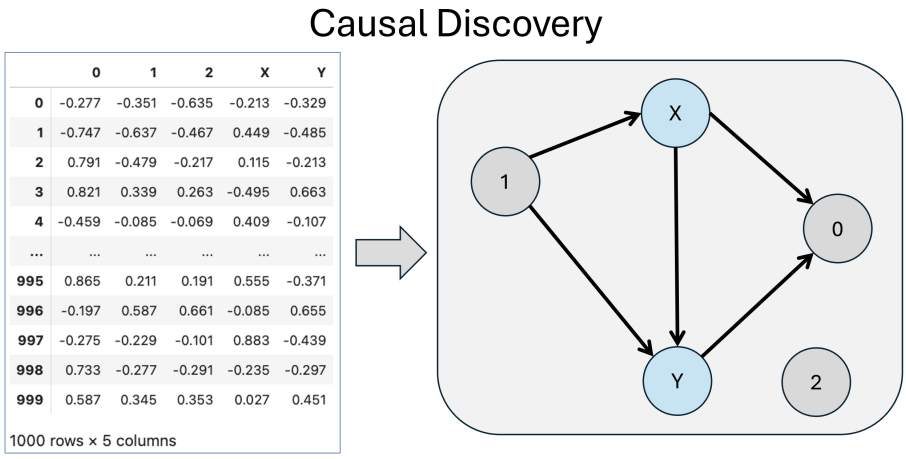
Abstract

This paper describes the ADIA Lab Causal Discovery Challenge conducted in 2024. The challenge focused on causal discovery, the task of inferring causal relationships from observational data, with an emphasis on the classification of variables according to their causal roles. Participants were tasked with identifying eight causal categories (Confounder, Collider, Mediator, Independent, Cause of X, Consequence of X, Cause of Y, and Consequence of Y) using a large synthetic database generated using known causal graphs. The competition attracted significant global participation, with 1,904 registered users submitting 3,343 solution attempts. We analyze the methodological approaches of the top-performing solutions, which achieved multiclass balanced accuracies of up to 76.70%, substantially outperforming baseline solutions (approximately 40%). An analysis of the winning methodologies showed that supervised learning approaches outperformed traditional constraint-based methods; sophisticated feature engineering and representation learning captured complex causal structures; ensemble methods combining predictions from multiple approaches yielded more robust results; and performance varied significantly across different graph structures, with hierarchical graphs being consistently easier to analyze than preferential attachment graphs. This report documents these findings and explores their implications for advancing causal discovery research.

1 Introduction

Much of scientific inquiry is based on the generation or repurposing of data to make causal inferences. Scientists who make causal inferences quantify the magnitude and precision of causal effects by combining data with assumptions about the causal structure of the world Rubin [1974], Robins [1986]. These causal assumptions, or expert knowledge, can be represented as causal directed acyclic graphs (DAGs) with nodes that represent variables and directed edges that indicate causal direct effects Pearl [2009], Spirtes et al. [2000], Greenland et al. [1999]. Existing methods for causal inference, which rely on a variety of causal assumptions Hernán and Robins [2020], Imbens and Rubin [2015], are applied to estimate the effects of interventions in the health sciences, economics, climate sciences (e.g., Runge et al. [2019]), investments López de Prado [2023], López de Prado et al. [2024, 2025], López de Prado and Zoonekynd [2026], and many other fields.

Rather than using causal DAGs as a way to represent our assumptions or expert knowledge about the causal structure, we can attempt to use the data to learn the causal structure without making assumptions about the causal structure. The process of learning components of the



	0	1	2	X	Y
0	-0.277	-0.351	-0.635	-0.213	-0.329
1	-0.747	-0.637	-0.467	0.449	-0.485
2	0.791	-0.479	-0.217	0.115	-0.213
3	0.821	0.339	0.263	-0.495	0.663
4	-0.459	-0.085	-0.069	0.409	-0.107
...	...	...	...	...	...
995	0.865	0.211	0.191	0.555	-0.371
996	-0.197	0.587	0.661	-0.085	0.655
997	-0.275	-0.229	-0.101	0.883	-0.439
998	0.733	-0.277	-0.291	-0.235	-0.297
999	0.587	0.345	0.353	0.027	0.451

1000 rows × 5 columns

Figure 1: The fundamental causal discovery problem: inferring a causal DAG from observational data.

causal structure through data analysis is referred to as *causal discovery*, represented in Figure 1. See the books by Spirtes et al. [2000] and by Peters et al. [2017] for descriptions of approaches to causal discovery.

The methodological landscape of causal discovery has evolved substantially over recent decades. Classical constraint-based methods, such as the Peter-Clark (PC) algorithm, begin by establishing an undirected skeleton through conditional independence testing before orienting edges based on specific patterns Kalisch and Bühlmann [2007]. Score-based approaches search through the space of possible DAGs to optimize scoring functions that balance model fit against complexity Tsamardinos et al. [2006]. More recently, the field has witnessed the emergence of hybrid methods that combine elements of both approaches Glymour et al. [2019], and machine learning techniques that leverage advances in computational capacity and algorithm design Ng et al. [2020].

Despite this methodological progress, causal discovery faces several critical challenges in practice. First, causal discovery must assume faithfulness (statistical independencies in the observed data imply missing causal arrows on the DAG) and, generally, knowledge about the time sequence of the variables on the DAG. Second, in the presence of unmeasured confounders (common causes of treatment and outcome), causal discovery is often impossible. Third, even in cases when causal discovery is possible, the fact that data samples are finite implies that the validity of causal discovery from observational data relies heavily on a priori subject-matter knowledge about the plausibility of various causal diagrams.

A special setting in which causal discovery could be manageable is one in which 1) the true causal DAG (the ground truth) is known for a large number of training datasets, and 2) researchers want to discover the causal DAG in a new dataset with a data generation mechanism similar to that of the training datasets. In this setting, causal discovery algorithms do not have to rely exclusively on statistical independencies in the new dataset; rather, they can also identify data patterns (using supervised learning approaches) for different DAGs in the training datasets and use information on those patterns to propose a causal structure for the new dataset.

Using this special setting, ADIA Lab launched the Causal Discovery Challenge in 2024 ADIALab [2024]. This global competition invited researchers and practitioners to develop novel algorithms for causal discovery that can take advantage of a large number of known dataset-DAG pairs. The emphasis was not unconstrained edge detection, but the correct classification of nodes other than the treatment (X) and the outcome (Y) into confounders, mediators, colliders, and other causal categories.

The eight causal relationship categories used in the ADIA Lab Challenge are: Confounder,

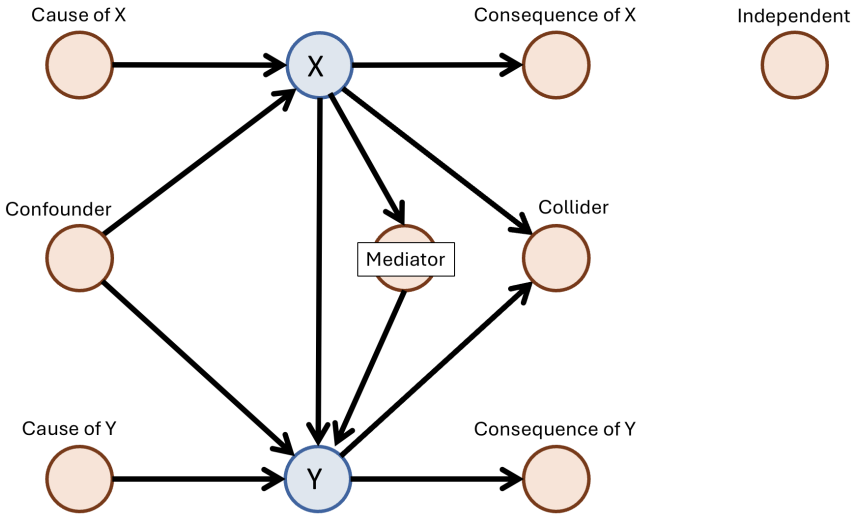


Figure 2: The eight causal relationship categories used in the ADIA Lab Challenge: Confounder, Collider, Mediator, Independent, Cause of X, Consequence of X, Cause of Y, and Consequence of Y.

Collider, Mediator, Independent, Cause of X, Consequence of X, Cause of Y, and Consequence of Y. We only look at the 1-hop neighbourhood of X and Y.

In the following sections, we provide an overview of the competition structure, datasets, evaluation metrics, and baseline solutions provided by ADIA Lab. We then present an analysis of the participation patterns and results, with particular attention to the methodologies employed by top-performing teams. Finally, we discuss the implications of these findings for causal discovery and identify directions for future research.

2 Materials and Methods

2.1 Datasets

The ADIA Lab Causal Discovery Challenge utilized a collection of 47,000 synthetic datasets, each containing about 1,000 observations of 3-10 variables . The methodology for generating these synthetic datasets is described in Appendix: Database Generation. The datasets were organized in two groups: a training database and a test one, as illustrated in Figure 3.

2.1.1 Training Database

The training database consisted of approximately 25,000 datasets. Each training dataset included:

1. A tabular data file containing approximately 1,000 observations of 3-10 numerical variables, two of which were identified as treatment X and outcome Y.

2. The corresponding (ground truth) causal DAG that generated the data.

Each variable in the dataset, other than the designated treatment X and outcome Y variables, belonged to one of the 8 causal relationship categories illustrated in Figure 2. These categories (Confounder, Collider, Mediator, Independent, Cause of X, Consequence of X, Cause of Y, and Consequence of Y) capture key causal relations to treatment and outcome.

2.1.2 Test Database

The test database comprised approximately 22,000 datasets with structures similar to those in the training database but without provided ground truth causal graphs. The test database was divided into:

1. **Public Test Set**: Used for immediate feedback and leaderboard ranking throughout the competition.
2. **Private Test Set**: Reserved for final evaluation and determination of winners.

This two-part evaluation strategy helped prevent overfitting to the public leaderboard while ensuring that winning solutions demonstrated generalizable performance on previously unseen data.

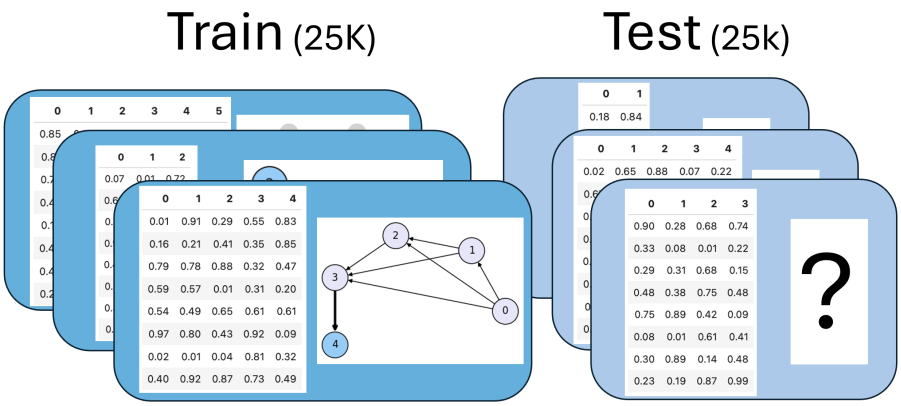


Figure 3: Illustration of the training datasets with ground truth DAGs (left) and test datasets requiring prediction (right).

In Figure 3 we illustrate the training datasets with ground truth DAGs (left) and test datasets requiring prediction (right).

Both the training and test datasets represented a wide range of causal scenarios, with varying numbers of variables, causal relationship categories, and statistical dependencies. This diversity ensured that successful algorithms would need to handle several types of causal structures.

2.2 Evaluation Metric

In the competition, the submitted algorithms attempted to identify aspects of the causal graph of a dataset. As the evaluation metric, we computed the probability of each algorithm correctly identifying the eight causal relationship categories for the variables in the graph. The average of these eight probabilities – referred to as category accuracies – served as the competition's scoring metric: the *multiclass balanced accuracy* Kelleher et al. [2020], Guyon et al. [2015], also known as *average per-class accuracy*. For example, for the category "confounder," we calculated the probability of correctly identifying a variable as a confounder. This procedure was repeated for each category, and the resulting eight values were averaged.

On average, each dataset in the test database included five variables, excluding X and Y. Therefore, the total number of variables in the competition's test database exceeded 100,000, distributed across the eight categories. The multiclass balanced accuracy is defined as:

$$multiclass\ balanced\ accuracy = \frac{1}{8}\sum_{i=1}^{8}\frac{TP_i + TN_i}{TP_i + FP_i + TN_i + FN_i}$$

where TP_i represents the true positives, FP_i the false positives, TN_i the true negatives, and FN_i for category i, respectively.

Across the entire test database, each category was roughly equally represented. Nevertheless, the multiclass balanced accuracy is robust to class imbalances. Consequently, a random guessing strategy would achieve a multiclass balanced accuracy of 12.5% (1/8), providing a baseline for algorithm performance. Final rankings were determined based on performance on the private test set, with participants' submissions evaluated using this metric, as implemented in the Python package scikit-learn[1].

2.3 Baseline Solutions Provided by ADIA Lab

To establish performance benchmarks and provide starting points for participants, we developed and shared three baseline solutions representing different methodological approaches to causal discovery. The implementations of these baselines were made available to participants as Jupyter Notebooks on GitHub CrunchDAO [2024].

2.3.1 Basic PC Algorithm

The first baseline solution used a classical constraint-based approach. The Peter-Clark (PC) algorithm Chickering [2002] starts with a fully connected undirected graph, eliminates edges based on conditional independence tests, and orients remaining edges using rules that identify presence of absence of v-structures. Therefore, the output of the PC algorithm is not a DAG, because the direction of some edges is not known. Algorithms that output a directed but not acyclic graph can be *fixed* by solving an optimization problem where the objective was to find the closest DAG to a given directed graph by minimizing a weighted sum of penalties for flipped, removed, and added edges.

The implementation of this baseline method, using the CASTLE (Causal Structure Learning) library[2], is available at:

- `https://github.com/crunchdao/competitions/blob/master/competitions/causality-discovery/quickstarters/baseline/baseline.ipynb`

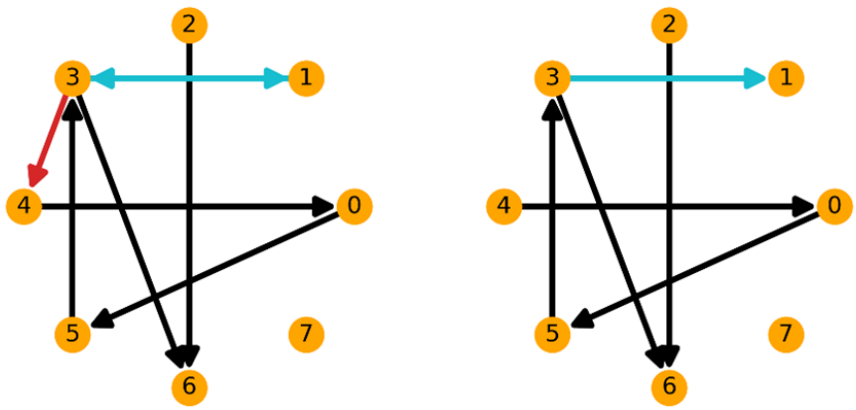


Figure 4: Illustration of cycle removal in a directed graph (left) to create a directed acyclic graph (right).

In Figure 4 we illustrate cycle removal in a directed graph (left) to create a directed acyclic graph (right).

[1] `https://scikit-learn.org/stable/modules/model_evaluation.html#balanced-accuracy-score`
[2] `https://github.com/huawei-noah/trustworthyAI`

2.3.2 Traditional Machine Learning

The second baseline converted the causal discovery problem into a supervised machine learning task by:

1. Extracting features from variables in the training datasets, including their relations with X and Y via correlations, t-statistics, and other associational measures.
2. Creating vector embeddings for each variable based on a 13-dimensional feature space.
3. Training a RandomForest classifier to predict each of the eight causal relationship categories based on these embeddings. individual variable classifications are reconstructed into complete causal graphs by converting each relationship category into the appropriate adjacency matrix entries.

The implementation of the Random Forest classifier, with 100 estimators and balanced class weights to handle category imbalances, together with a tutorial explanation is available at:

- `https://github.com/crunchdao/competitions/blob/master/competitions/causality-discovery/quickstarters/supervised_baseline/supervised_baseline.ipynb`

2.3.3 Deep Learning

The third baseline solution employed a neural network approach to directly predict adjacency matrices representing causal graphs from observational data. This method included:

1. A transformer neural network that transforms input datasets of shape $(N \times M)$ into $(M \times M)$ adjacency matrices, where N is number of observations and M is number of variables.
2. An architecture based on two input layers that map the $(N \times M)$ input dataset into embeddings whose output $(M \times M \times 64)$ is processed by two final layers to produce $(M \times M)$ edge probabilities.
3. Transformation of latent representations into queries and keys, followed by the application of dot-product operations across different variables to form latent representations of the edges between variables.
4. Output layers that predicted directed edges based on these latent edge representations.

The implementation of this approach with descriptions of the neural network architecture is available at:

- `https://github.com/crunchdao/competitions/blob/master/competitions/causality-discovery/quickstarters/neural_network/neural_network.ipynb`

Each baseline solution achieved approximately 40% multiclass balanced accuracy, substantially outperforming random guessing (12.5%). The performance details and impact of these baselines are further discussed in the Section 3.2.

2.4 Competition Structure

The ADIA Lab Causal Discovery Challenge ran from August 1 to October 24, 2024, with an award ceremony held on November 15, 2024 (ADIA Lab, 2024), at the ADIA Lab Symposium. The competition offered a prize pool of $100,000, distributed among the top ten performing teams: $40,000 for first place, $20,000 for second place, $10,000 for third place, $5,000 each for fourth through seventh places, $3,500 each for eighth and ninth places, and $3,000 for tenth place.

The competition was hosted on the Crunch DAO platform[3], which facilitated participant registration, dataset access, submission evaluation, and leaderboard tracking. Participants received immediate feedback through a public leaderboard displaying performance metrics on a subset of test data, enabling iterative improvement throughout the competition period.

Participants were required to submit their solutions in the form of Python code (either as Jupyter notebooks or Python scripts) to the Crunch DAO platform. The submission procedure involved:

1. Submitting code solutions through a specific API provided by the platform.
2. The platform executed participants' code to both train their models on the training datasets and generate predictions on the test datasets.
3. Test datasets (both public and private) were never directly shared with the participants.
4. The model was trained in the cloud on the competition platform, and the platform then used the trained model to predict the causal relationships on the test set.
5. Code submissions needed to be computationally efficient, with runtime constraints to ensure feasible evaluation.
6. Submissions were evaluated automatically, with immediate feedback on public test set performance.
7. Participants could make multiple submissions throughout the competition period, with only their best-performing submission counting toward the final ranking.

This standardized submission procedure ensured fair evaluation across all participants while supporting the iterative development and refinement of algorithms throughout the competition.

2.5 Statistical Analysis

During the competition, only the overall multiclass balanced accuracy of each submission was presented on the leaderboard. After the competition conclusion, additional analyses were performed using the private test set results. These analyses included:

1. Overall multiclass balanced accuracy for each submission.
2. Per-category accuracy metrics to identify strengths and weaknesses in detecting specific causal relationship categories.
3. Comparative analysis between different methodological approaches.

Additionally, post-competition analysis included an LLM- (large language model) based clustering of submission approaches to identify methodological patterns among participants.

[3] `https://hub.crunchdao.com/competitions/causality-discovery`

3 Results

3.1 Participation and Submission Overview

The ADIA Lab Causal Discovery Challenge attracted significant global interest from the research community. By the competition's conclusion, a total of 1,904 users had registered on the platform, with participation steadily increasing throughout the competition period (Figure 5). The geographical distribution of participants was broad, with large numbers of active users in the United States, India, China, and western Europe (Figure 6).

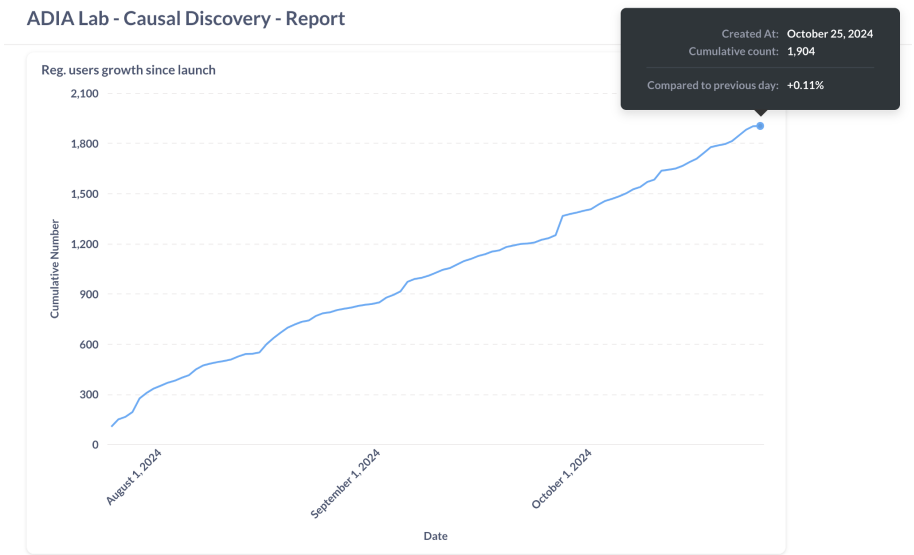


Figure 5: Cumulative number of registered users over time.

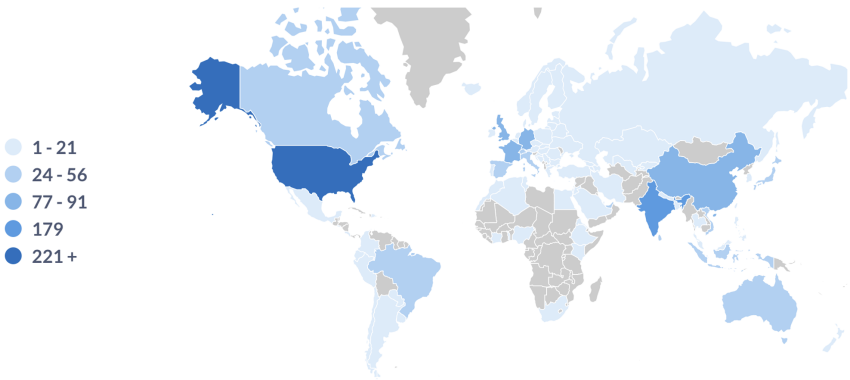


Figure 6: Geographical provenance of the most active users.

Participants collectively submitted 3,343 solution attempts by the end of the competition (Figure 7). Submission activity followed a consistent upward trend, with noticeable acceleration in the final weeks as participants refined their approaches.

3.2 Performance of Baseline Solutions

The three baseline solutions (constraint-based PC algorithm, traditional supervised learning, and deep learning) achieved comparable performance, with approximately 40% multiclass bal-

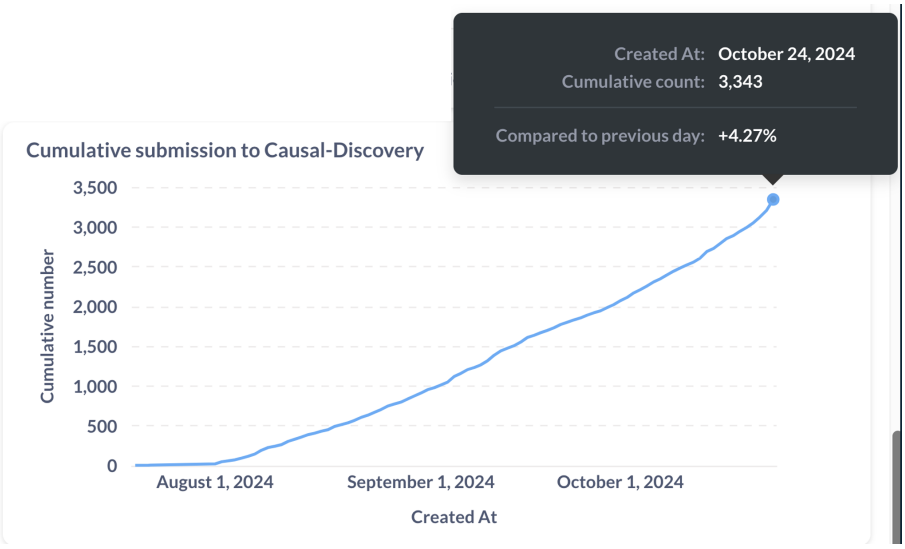


Figure 7: Cumulative number of submissions over time.

anced accuracy across the eight causal relationship categories. Post hoc analysis of per-class accuracy metrics (as shown in Table 1) showed that the performance of traditional supervised learning varied across relationship categories, ranging from greater than 50% for "Cause of Y" (62.51%), "Independent" (56.12%), and "Cause of X" (50.88%), to less than 20% for "Mediator" (13.89%) and "Confounder" (18.60%) relationships.

3.3 Performance of Winning Solutions

The winning solutions in the ADIA Lab Causal Discovery Challenge demonstrated notable improvements over the baseline approaches. The first-place solution by Hicham Hallak achieved a multiclass balanced accuracy of 76.70%, with the top three solutions all exceeding 74% multiclass balanced accuracy.

Table 1 presents the performance metrics of the top 10 winning solutions: the overall multiclass balanced accuracy and per-category accuracy percentages calculated after the competition.

Several key patterns emerge from the comparison of these winning solutions with the baseline solutions:

1. Overall, all top solutions significantly outperformed the baseline solutions, with even the 10th-place solution achieving a 66.34% accuracy compared with 40% for the baseline solutions.

2. The improvements of top-performing solutions over baseline solutions were even greater for some causal categories. For example, the first-place solution achieved 73.19% accuracy on confounders (compared to 18.60% for the supervised baseline) and 70.17% on colliders.

3. Across all top solutions, the "Independent" and "Cause of Y" relationship categories had the highest accuracy, while "Mediator" relationships had the lowest.

The winning solutions built upon the supervised learning approach established by the best baseline but incorporated significant innovations in their techniques to capture nuanced causal relationships. The top performers used diverse approaches including GNN- (graph neural network) inspired edge-feature architectures (1st place), extensive feature engineering with XGBoost ensembles (3rd place), k-nearest neighbor conditional mutual information features with LGBM (4th place), and comprehensive feature sets with LGBM and AutoML integration (5th place).

Table 1: Post hoc performance metrics of the top 10 winning solutions, showing overall multiclass balanced accuracy and per-category accuracy percentages calculated after the competition.

Pos	Winner(s)	Avg. Score (%)	Conf	Coll	Med	Cause X	Cause Y	Conseq X	Conseq Y	Indep
1	Hicham Hallak	76.70%	73.19	70.17	65.67	79.35	81.68	76.53	80.80	86.18
2	Andre Franca & Alexis Gassmann	74.47%	71.91	72.52	65.44	70.28	84.83	69.56	72.66	88.57
3	Mutian Hong & Guoqin Gu	74.06%	63.91	66.91	63.91	75.62	82.52	74.65	80.41	84.51
4	Alexander Kiechle-Cornish	72.88%	70.50	63.07	53.99	77.97	82.40	72.36	77.65	85.10
5	Hoàng Thiên Nữr	70.37%	63.89	54.99	50.93	68.96	84.26	75.32	74.83	89.77
6	Jorge Linde Díaz	70.09%	74.36	64.59	63.83	63.39	84.39	65.88	60.94	83.31
7	Bao Duong	69.11%	62.29	50.99	47.82	68.54	83.40	76.14	71.78	91.91
8	Grzegorz Tratkowski	68.62%	65.19	57.37	58.73	73.36	76.02	65.18	76.10	77.05
9	Quang-Duy Tran	66.59%	58.09	54.71	48.60	70.30	78.21	65.76	74.45	82.60
10	Thin Nguyen	66.34%	60.35	49.50	40.57	68.79	81.07	73.52	68.83	88.08
	Supervised Baseline	39.68%	18.60	43.65	13.89	50.88	62.51	26.74	45.04	56.12

3.4 Performance by Graph Generator Type

We examined how the top solutions performed across the different graph generator types: six synthetic graph models (Hierarchical, Bipartite, Erdős-Rényi, Low-rank, Preferential Attachment, Scale-free) plus samples from a real-world network. Figure tab:performance-by-generator presents the multiclass balanced accuracy for each winner broken down by graph generator type.

In Table 2 we show the performance of the top 10 winning solutions across different graph generator types.

Hierarchical graphs were the easiest to analyze across all solutions, whereas **Preferential Attachment models** were the most difficult. Because the **relative performance gap** between different graph generators remains fairly consistent across competitors, these difficulty differences seem to be inherent to the graph structures rather than specific to particular algorithms.

The **first-place solution** showed more consistent performance across graph types than other competitors, with less variance between the highest and lowest scores. This suggests superior robustness to variations in graph structure, an important consideration for real-world applications.

3.5 LLM-based Post hoc Analysis

We analyzed all submitted solutions using a large language model (LLM) to identify clusters and patterns in the approaches. More precisely, we extracted the code of all the submissions, computed an embedding of each of them with a LLM trained on code, applied UMAP dimension reduction, computed clusters with DBScan, and plotted the data with datamapplot McInnes et al. [2023]. This analysis revealed (Figure 8) several methodological clusters of submitted solutions, including neural networks, traditional supervised learning algorithms with various enhancements (e.g., manual feature engineering), and classical causal discovery algorithms (e.g., constraint-based PC algorithm, GES (greedy equivalence search), LiNGAM linear non-Gaussian additive model). A smaller cluster employed ensemble methods starting with the AVICI algorithm Lorch et al. [2022].

Figure 8 shows the LLM-based clustering of solution approaches.

This analysis also revealed a clear relationship between methodological approach and performance, with solutions in the "Neural Networks" and "Traditional Supervised learning" clusters achieving the highest average accuracies. The visualization of high-scoring solutions (Figure 9) shows a concentration of successful approaches in these methodological clusters, particularly those scoring above 50% accuracy.

Figure 9 show the submissions map where score is greater than 0.5.

This analysis demonstrates the influence of the baseline solutions on participants' approaches, with many high-performing submissions drawing inspiration from the three baseline solutions (particularly extending the supervised learning framework). The top solutions, however, diverged significantly from the baselines in their implementation details and algorithmic innovations.

4 Discussion

The ADIA Lab Causal Discovery Challenge was conducted using synthetic data with known ground truth causal structures, a special case that may not directly translate to typical real-world applications. In particular, the competition datasets were generated with no unmeasured confounders, which makes causal discovery more manageable.

Working under these constraints, the Challenge enabled objective evaluation and comparison of different types of algorithms. Specifically, it shed light on the relative performance of supervised learning approaches when causal discovery can be transformed into a classification

Table 2: Performance of the top 10 winning solutions across different graph generator types.

Pos	Winner(s)	Bipartite	Hierarchical	Sample	Erdos-Renyi	Low-rank	Scale-free	Pref.Attach.
1	Hicham Hallak	79.65	84.21	77.20	79.56	78.59	79.21	72.12
2	Andre Franca & Alexis Gassmann	78.61	80.88	76.51	79.33	76.43	81.16	69.15
3	Mutian Hong & Guoqin Gu	78.52	81.53	75.29	69.93	71.25	73.04	69.24
4	Alexander Kiechle-Cornish	78.87	76.26	73.78	73.02	71.05	72.44	68.25
5	Hoàng Thiên Nữr	77.00	75.78	69.85	69.73	67.68	67.26	67.93
6	Jorge Linde Díaz	68.93	69.71	72.38	68.06	67.33	68.11	64.17
7	Bao Duong	78.26	74.75	68.14	69.99	68.03	66.81	67.00
8	Grzegorz Tratkowski	70.90	76.17	69.22	68.86	70.24	66.85	62.63
9	Quang-Duy Tran	69.14	74.10	66.66	66.72	66.54	64.90	63.00
10	Thin Nguyen	74.09	70.27	66.47	61.39	61.61	61.90	63.61
	Supervised Baseline	40.95	41.55	41.15	39.23	40.99	38.62	34.17

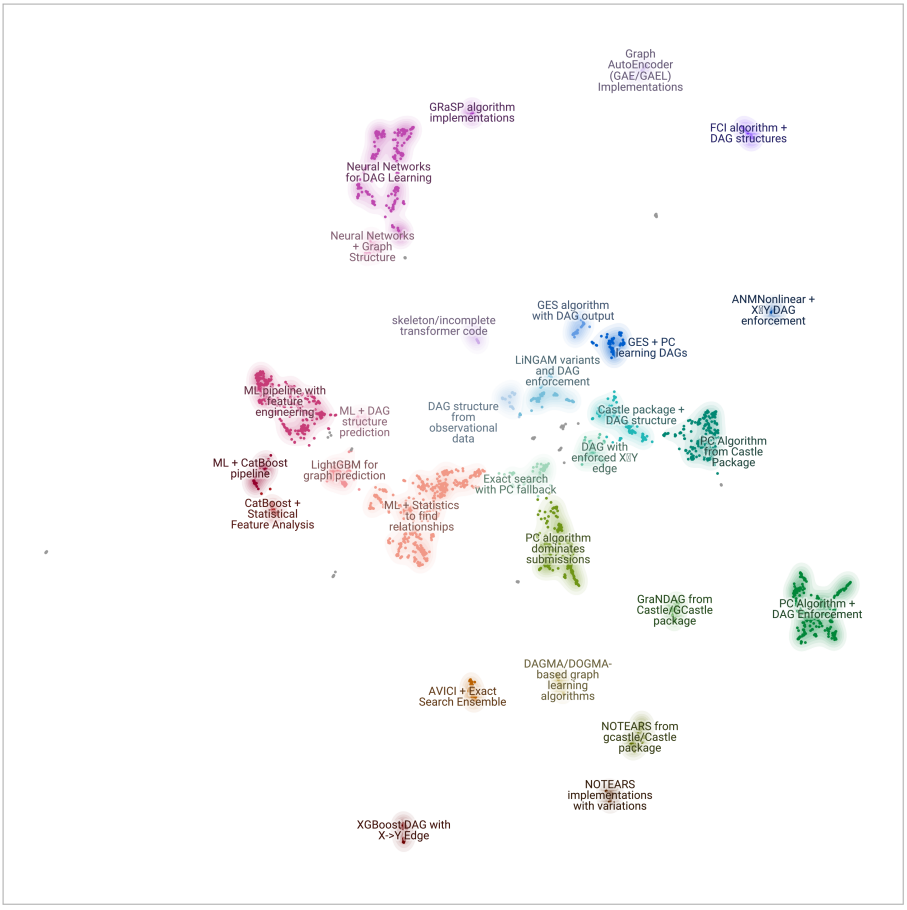


Figure 8: LLM-based clustering of solution approaches.

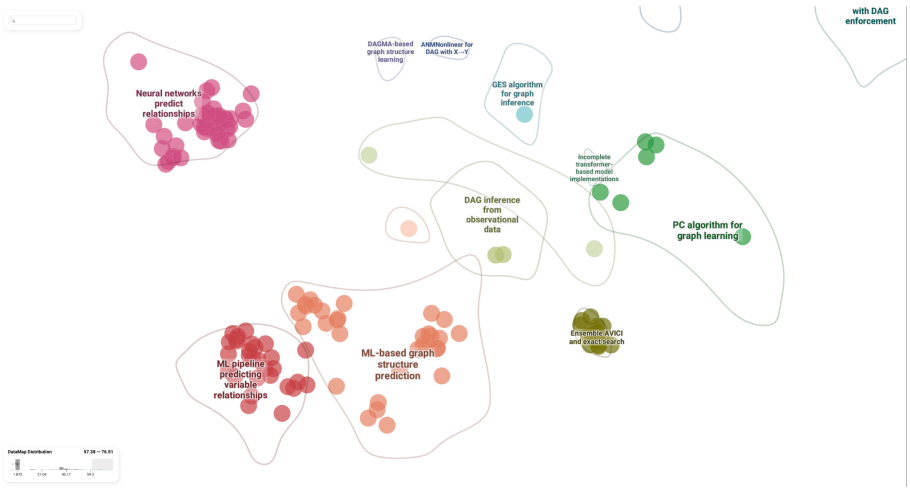


Figure 9: Submissions map where score is greater than 0.5.

problem. This section discusses the winning methodologies and identifies remaining challenges and opportunities.

4.1 Characteristics of Winning Approaches

4.1.1 Supervised Learning Algorithms

Supervised learning approaches recognize patterns associated with specific causal relationships from the training data. The top-performing solution utilized an approach inspired by Graph Neural Networks (GNNs) that focused on the relationships between variables (edges in the causal graph) to determine the causal roles of each variable. This solution demonstrated remarkable consistency across different graph structures and graph types.

4.1.2 Integration of Supervised and Unsupervised Methodologies

Many of the top solutions were hybrid approaches that integrated the pattern recognition capabilities of supervised models with the information provided by traditional causal discovery algorithms. The top-performing solution utilized an approach inspired by Graph Neural Networks (GNNs) that focused on the relationships between variables (edges in the causal graph). This solution leveraged adjacency prediction during training, but at inference time relied solely on the node (variable) classification, similar to the supervised baseline. The fourth-place solution extracted multiple feature sets including k-nearest neighbour conditional mutual information features, used a tweaked version of the baseline solution to extract additional features, and then used probability-weighted predictions as features, along with features from GES, BIC (Bayesian Information Criterion) Exact Search, and DirectLiNGAM, before applying supervised classification via Light Gradient Boosting Machine (LGBM). The third-place solution primarily relied on extensive feature engineering combined with boosted tree models and included outputs from various causal discovery algorithms (ExactSearch, PC, FCI (fast causal inference), GRaSP (greedy relaxed skeleton pruning), DML (double machine learning)) as just one part of their feature set.

Rather than viewing supervised and unsupervised approaches as competing paradigms, the competition results indicate that they can be complementary, each addressing different aspects of the causal discovery challenge.

4.1.3 Innovations in Feature Engineering and Representation Learning

Feature engineering and representation learning in the top solutions were based on rich, multi-dimensional representations of variable relationships that captured both direct interactions and higher-order dependencies.

The first-place solution focused on pairwise relationships between variables (rather than on node characteristics). The third-place solution employed extensive feature engineering including correlation measures, information-theoretic features, discretization-based statistical features, and regression coefficients. The fifth-place solution created comprehensive statistical features categorized into several groups including ANM- (additive noise model) based features, correlation-based features, and conditional independence metrics. These approaches allowed the models to capture subtle patterns that distinguished between similar causal structures, such as confounders and mediators. In contrast, the baseline approaches struggled to identify these relationship categories.

4.1.4 Ensemble Methods and Model Combination

Another common pattern among the top-performing solutions was the use of ensemble methods and model combination techniques. Rather than relying on a single algorithm or model, many

of the top competitors integrated predictions from multiple approaches.

This ensemble approach was evident in various forms across the top solutions. The second-place solution employed an ensemble of AVICI Lorch et al. [2022] models combined with traditional causal discovery algorithms, incorporating exact structure learning, a hybrid A search algorithm, and sophisticated cycle removal procedures. Mutian Hong and Guoqin Gu's third-place solution created an ensemble of three XGBoost models with different numbers of estimators (7,000, 7,500, and 8,000). Hoàng Thiên Nữr's fifth-place solution initially used a hand-tuned LGBM model, but in the final stages incorporated AutoGluon (AutoML), which outperformed the carefully tuned LGBM model on its own, achieving 69% accuracy. This strategy of combining different models proved particularly effective at handling the diversity of causal structures present in the competition datasets, as different models could specialize in recognizing different patterns.

The success of these ensemble approaches highlights the inherent complexity of causal discovery and suggests that no single algorithmic approach is likely to be optimal across all scenarios. Instead, the integration of multiple complementary methods appears to be a more promising strategy for achieving high accuracy on diverse datasets.

4.2 Future Challenges

The competition results suggest that when sufficient labeled training data is available, supervised approaches can outperform traditional constraint-based methods. However, four important challenges remain:

1. **Generalizability beyond synthetic data**: The competition utilized synthetically generated datasets without unmeasured variables and with structures that may not fully capture the complexities and nuances of real-world data. Further research is needed for more complex data generation mechanisms, including time-varying variables, unmeasured confounders and measurement error.

2. **Calibration of graph generators:** Evaluating causal discovery algorithms across a wide range of graph structures requires careful calibration of the underlying graph generation processes. Properly tuning the complexity and characteristics of the generated datasets and graphs is essential to ensure fair comparisons and to minimize systematic biases. This remains an active area of research.

3. **Computational efficiency**: Some of the top-performing approaches involved computationally intensive techniques that may be prohibitive for very large datasets or real-time applications. Research into more efficient algorithms that maintain high accuracy would increase practical utility.

4. **Interpretability**: While many of the top solutions achieved high accuracy, their complex, ensemble-based nature may limit interpretability. For domains where understanding the reasoning behind causal discoveries is crucial, developing more interpretable approaches without sacrificing performance remains an important goal.

5 Conclusion

The ADIA Lab Causal Discovery Challenge proposed a simplified setting that allows the integration of state-of-the-art methodologies. The solutions, which approached 80% accuracy, highlighted the value of supervised learning approaches when labeled training datasets are available, while also emphasizing the importance of incorporating traditional causal principles and constraints.

We view this competition as ADIA lab's first step to bring together investigators in causal discovery. The remaining challenges – including applicability to real-world data, computational efficiency, and interpretability – provide a roadmap for future research efforts and competitions.

6 Acknowledgements

We extend our sincere gratitude to the top five winners of the ADIA Lab Causal Discovery Challenge for their valuable feedback on this manuscript: Hicham Hallak (first place), Andre Franca and Alexis Gassmann (second place), Mutian Hong and Guoqin Gu (third place), Alexander Kiechle-Cornish (fourth place), and Hoàng Thiên Nữr (fifth place). Their insights and methodological innovations have significantly enriched this report and advanced the field of causal discovery.

We also wish to acknowledge the CrunchDAO platform for their fundamental role in the success of this competition. Their robust infrastructure supported 1,904 registered users who submitted 3,343 solution attempts from around the world, creating a truly global collaborative environment for advancing causal discovery research. The platform's capabilities for managing participant registration, dataset access, submission evaluation, and leaderboard tracking were instrumental in creating an effective competitive framework that fostered innovation in this challenging domain.

A Appendix

B Appendix: Top Solutions

This appendix provides detailed descriptions of the methodologies employed by the top-performing solutions in the ADIA Lab Causal Discovery Challenge. The approaches described here represent significant innovations in causal discovery methodology and provide valuable insights for future research in this field.

B.1 First Place Solution (76.70% Accuracy)

B.1.1 Hicham Hallak

Hicham Hallak's winning solution achieved a multiclass balanced accuracy of 76.70%, significantly outperforming the competition baselines, which all achieved approximately 40% accuracy. The solution employed an innovative Graph Neural Network (GNN) inspired approach that used edge features to classify nodes, a significant departure from conventional methods.

As described by Hallak himself: "In a nutshell, the solution can be seen as a Graph Neural Network where edge features are used to classify nodes. With GNNs, node classification is often performed by leveraging node data to generate node embeddings. Sometimes, edge data are also incorporated to improve overall performance. However, here, we exclusively rely on edge features to classify nodes."

This approach was particularly novel because instead of using node characteristics directly, it focused on the relationships between variables (edges in the causal graph) to determine the causal roles of each variable (nodes). This methodological innovation allowed the solution to effectively capture the complex dependencies required for accurate causal role classification.

Key components of the winning solution included:

1. **Edge-Feature Focus**: Rather than directly extracting features from individual variables, the approach focused on pairwise relationships between variables, extracting features that characterized the potential causal connections between them.

2. **GNN-Inspired Architecture**: The solution implemented a graph-based learning approach that aggregated edge information to classify nodes according to their causal roles in the network.

3. **Feature Engineering**: Comprehensive statistical features were generated to characterize the relationships between variables, capturing both direct and indirect causal effects.

4. **Dual Training Approach**: The solution used an adjacency prediction branch during training that helped improve model performance, but this branch was not used during inference.

5. **Node Classification Focus**: At inference time, only the node (variable) classification head was used (similar to the supervised baseline), which effectively distinguished between subtle causal relationships like confounders and mediators without requiring explicit causal consistency enforcement.

What made this solution particularly effective was its departure from traditional node classification methods, instead leveraging the rich information contained in the relationships between variables to determine their causal roles. This approach enabled the model to better capture the essence of causal discovery: understanding how variables influence each other within a network of relationships.

B.2 Second Place Solution (74.47% Accuracy)

B.2.1 Andre Franca & Alexis Gassmann

The second-place solution by Andre Franca and Alexis Gassmann achieved a multiclass balanced accuracy of 74.47%, just 2.23 percentage points behind the winning solution. Their approach combined ensemble methods from pre-trained deep learning models (AVICI) with traditional causal discovery algorithms.

Key components of their solution included:

1. **Ensemble of AVICI Models**: They utilized multiple pre-trained AVICI (Alternating Variable Conditioning for Score-based Causal Inference) Lorch et al. [2022] models, which had been trained to predict causal relationships directly from observational data. The predictions from these models were averaged to create robust edge probability estimates.

2. **Exact Structure Learning**: Their solution integrated an exact structure learning approach that leveraged polynomial feature transformations and Bayesian Information Criterion (BIC) scoring to evaluate potential graph structures.

3. **Hybrid A^* Search Algorithm**: They implemented a custom A^* search algorithm for finding the optimal directed acyclic graph structure, incorporating path extensions and heuristic evaluations to efficiently navigate the search space.

4. **Cycle Removal via Causal Reasoning**: A sophisticated cycle removal procedure identified and eliminated the weakest edges in cycles based on causal strength metrics while ensuring that important edges (particularly $X \to Y$) were preserved.

5. **Feature and Edge Thresholding**: The solution employed carefully tuned thresholding parameters for different edge types (from X, to X, from Y, to Y, and other relationships), optimized to achieve high accuracy across all relationship categories.

The solution was particularly effective at identifying independent relationships (88.57% accuracy) and collider relationships (72.52% accuracy), outperforming even the first-place solution in these categories. This effectiveness likely stems from the combination of deep learning models

that captured complex patterns in the data and traditional causal discovery algorithms that incorporated structural constraints.

B.3 Third Place Solution (74.06% Accuracy)

B.3.1 Mutian Hong & Guoqin Gu

Mutian Hong and Guoqin Gu secured the third position with a multiclass balanced accuracy of 74.06%. Their solution primarily relied on extensive feature engineering combined with boosted tree models for supervised training, building upon the framework provided in the official supervised baseline.

The key components of their approach included:

1. **Extensive Feature Engineering**: Most of their efforts were directed towards designing effective features, including correlation-based features (Pearson, Spearman, Kendall, distance, partial), information-theoretic features (mutual information, conditional mutual information, copula entropy), discretization-based statistical features, regression coefficients, and more. They also used causal discovery algorithm outputs (ExactSearch, PC, FCI, GRaSP, DML) as features rather than using them as the core method.

2. **Data Augmentation**: A significant innovation was their data augmentation strategy, which remapped variable names while keeping the causal graph unchanged. This approach increased their training data by up to 11.21 times and improved performance by approximately 5%.

3. **Supervised Learning Models**: For the final classification, they created an ensemble of three XGBoost models with different numbers of estimators (7,000, 7,500, and 8,000) but otherwise identical hyperparameters. Initially, they used Random Forest for rapid iterations, later switching to LGBM for further evaluations.

4. **Categorical Feature Processing**: Variables with fewer than 10 unique values were treated as categorical features and encoded appropriately.

The third-place solution demonstrated balanced performance across most relationship categories, with particular strength in identifying causes of X (75.62%) and consequences of Y (80.41%). Their approach shows that with sophisticated feature engineering and ensemble methods, supervised learning can effectively tackle causal discovery challenges.

B.4 Fourth Place Solution (72.88% Accuracy)

B.4.1 Alexander Kiechle-Cornish

Alexander Kiechle-Cornish's fourth-place solution achieved a multiclass balanced accuracy of 72.88%. This approach integrated both supervised and unsupervised causal discovery methods, focusing on feature extraction from observational data.

The core components of this solution were:

1. **Feature Extraction**: Kiechle-Cornish extracted multiple feature sets including k-nearest neighbour conditional mutual information (CMI) features, capturing complex relationships between variables.

2. **Feature Generation from Models**: in this solution, a tweaked version of the Deep Learning baseline is used to extract features for classification. Additional features were extracted both from outputs of a neural network (attention-based model trained during the challenge) and and from unsupervised causal discovery tools (GES, BIC exact search, DirectLiNGAM).

3. **Machine Learning Classification**: The solution used Light Gradient Boosting Machine (LGBM, 3000 trees) to classify variables based on their causal relationships.

The fourth-place solution performed particularly well on cause-effect relationships, achieving 77.97% accuracy on "Cause of X" and 82.40% on "Cause of Y" categories, suggesting strong capability in determining causal direction.

B.4.2 Fifth Place Solution (70.37% Accuracy)

B.4.3 Hoàng Thiê Nũr

Hoàng Thiê Nũr secured fifth place with a multiclass balanced accuracy of 70.37%. Hoàng's approach focused on supervised learning with extensive feature engineering and model ensembling.

Key aspects of this solution included:

1. Feature Engineering: The approach involved creating a comprehensive set of statistical features categorized into several groups:
   - ANM-based features: Reconstructing underlying noise in Structural Equation Models between variables.
   - Correlation-based features: Pearson, Spearman, distance, and Kendall tau,
   - Conditional independence-based features: Both parametric and non-parametric metrics.
   - Information theory-based features: Related to entropy and mutual information.
   - Feature importance-based features: Evaluating variable roles using various machine learning models.
   - Causal features: Including mediation analysis of relationships like $X \to Y \to K$, $K \to X \to Y$, and $X \to K \to Y$.
2. **Model Optimization**: Initially using LGBM models (700 estimators, maximum depth of 15) with later adjustments to accommodate increasing feature complexity, including increased estimators, deeper tree structure, and reduced learning rate.
3. **Imbalanced Data Handling**: Employed SMOTE (Synthetic minority oversampling technique) oversampling and 5-fold cross-validation for robust model evaluation.
4. **AutoML Integration**: In the final stages, incorporated AutoGluon which outperformed carefully tuned LGBM models, achieving 69% accuracy.
5. **Final Ensemble**: The final solution combined the hand-tuned LGBM model with the AutoML model for optimal performance.

Hoàng's solution achieved remarkably similar scores between public (70.686%) and out-of-sample (70.37%) evaluations, demonstrating its robust generalization capability. The solution performed particularly well in identifying independent relationships and "Cause of Y" relationships.

B.5 Common Themes and Innovations Across Top Solutions

Several common themes and innovative approaches emerged across the top-performing solutions in the ADIA Lab Causal Discovery Challenge:

1. **Hybrid Methodologies**: All top solutions combined elements from multiple methodological traditions, including constraint-based methods, score-based approaches, and machine learning techniques. This hybridization allowed them to leverage the strengths of each approach while mitigating their individual weaknesses.

2. **Feature Engineering**: Extensive feature engineering was a critical component of successful solutions, with competitors developing sophisticated statistical and informational metrics to capture complex relationships between variables.

3. **Pairwise Relationship Focus**: Many top solutions approached the problem by first focusing on pairwise causal relationships between variables, then combining these insights to classify variables according to the eight causal relationship categories.

4. **Ensemble Methods**: The use of ensemble methods was widespread among top performers, combining predictions from multiple algorithms or models to improve robustness and accuracy.

5. **Structural Consistency Approaches**: Several successful solutions incorporated explicit constraints to ensure the logical consistency of the predicted causal structures, particularly the acyclicity requirement for directed graphs. Others, like Hallak's winning approach, achieved success without explicit causal consistency enforcement during inference.

These common elements highlight the importance of methodological diversity and integration in advancing the state of causal discovery. By combining insights from various disciplines and approaches, the top-performing solutions were able to achieve significant improvements over baseline methods, demonstrating the potential for continued progress in this challenging field.

C Appendix: Database Generation

C.1 Introduction

This appendix documents the methodology employed to generate the datasets for the ADIA Lab Causal Discovery Challenge. The generation process consisted of three interconnected components: (1) graph generation to create diverse causal structures, (2) random function generation to establish causal relationships between variables, and (3) dataset creation by combining these elements. This systematic approach ensured the creation of datasets that reflect a wide array of causal relationships and complexities, suitable for evaluating causal discovery algorithms.

C.2 Structural Causal Models

The foundation of the dataset generation process lies in structural causal models (SCMs), which define the data generation process. An SCM consists of a set of equations, each defining a random variable in terms of other variables and noise. The causal graph derived from an SCM encodes the relationships between variables: nodes represent random variables, and directed edges indicate causal influence. In the graph, an edge $X \to Y$ indicates that X is used in the definition of Y.

In the context of the competition datasets, the SCMs were designed to incorporate a range of functional relationships and noise levels, ensuring that the resulting datasets would challenge causal discovery algorithms with realistic complexity.

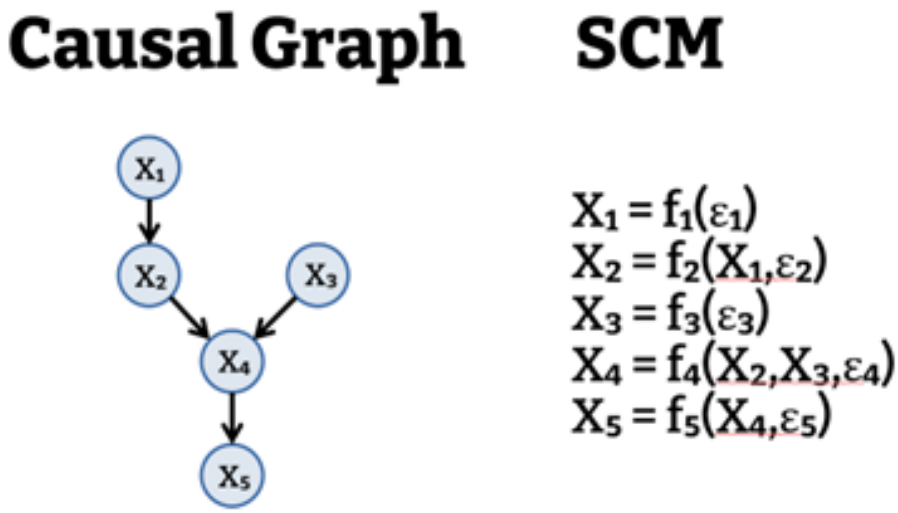


Figure 10: Relationship between a causal graph (left) and its corresponding structural causal model (right) with equations defining each variable.

C.3 Graph Generation

The first step in dataset creation involved generating diverse causal graph structures. The competition utilized both synthetic graph models and subgraphs derived from real-world networks to ensure broad coverage of possible causal structures.

C.3.1 Synthetic Graph Models

Six primary synthetic graph models were employed, each providing distinct structural characteristics:

1. **Preferential Attachment Model**: Nodes were added sequentially, with each new node connecting to 1–3 previously added nodes. This created graphs where earlier nodes tended to have higher degrees and more outgoing edges, mimicking the "rich get richer" phenomenon observed in many real networks.

2. **Erdős-Rényi Graphs**: Generated using the CASTLE library, these graphs contained 10 nodes and 15 edges distributed randomly according to a fixed probability of edge formation between any two nodes. This model produced graphs with a relatively uniform degree distribution.

3. **Scale-Free Graphs**: Also derived from the CASTLE library, these graphs comprised 10 nodes and 15 edges but followed a power-law degree distribution, where a small number of hub nodes had many connections while most nodes had few connections.

4. **Bipartite Graphs**: Nodes were divided into two distinct sets with a 1:1 split ratio, with edges only connecting nodes from different sets. The total configuration maintained 10 nodes and 15 edges.

5. **Hierarchical Graphs**: These structured graphs contained 10 nodes with an average degree of 3, organized into 5 hierarchical levels, reflecting organizational or taxonomic structures.

6. **Low-Rank Graphs**: Constructed with 10 nodes, an average degree of 2.5, and a rank of 5, these graphs exhibited specific structural complexity useful for testing algorithm robustness.

C.3.2 Real-World Graph Structures

To incorporate realistic causal structures, random induced subgraphs were extracted from the Messal Shale food web[4], a real-world directed graph that is nearly acyclic. The extraction process followed these steps:

1. Self-loops were removed from the original graph.
2. Only the largest connected component was retained.
3. A random node was selected as the starting point.
4. Up to three neighbors of this node were selected randomly.
5. The process was repeated for each selected neighbor's neighbors.

This methodology ensured the inclusion of naturally occurring causal structures alongside synthetic ones, enhancing the ecological validity of the dataset.

C.4 Random Function Generation

After establishing the graph structures, the next step involved defining the functional relationships between variables. Each node's value was computed as a (potentially noisy) function of its parent nodes' values. Three main approaches were used to generate these random functions:

C.4.1 Neural Networks

Simple neural networks with a single hidden layer were employed to create complex, non-linear relationships:

1. The hidden layer size was set to five times the input dimension (the number of parents).
2. Random weights were assigned to ensure variability.
3. ReLU (restricted linear unit) activation functions were applied.
4. The mapping function for each node followed the form: $x \mapsto Ax + a \mapsto \sigma(Ax + a) \mapsto B\sigma(Ax + a) + b$ where A is an $h \times k$ matrix, a is an $h \times 1$ vector, B is a $1 \times h$ matrix, b is a 1×1 matrix, and σ is the ReLU activation function.
5. Gaussian noise with standard deviations ranging from 0 to 1 was added to introduce stochasticity.

C.4.2 Bayesian Models

Bayesian models offered another approach to defining functional relationships:

1. Data was sampled from $F(\theta(x))$, where F was a probability distribution, $\theta(x)$ are the parameters, and x the parents of the node whose values y are sampled – for a Gaussian distribution, $\theta(x) = \big(\mu(x), \sigma(x)^2\big)$, and $y \sim N\big(\mu(x), \sigma(x)^2\big)$: this allows for heteroskedasticity.
2. The parameters $\theta(x)$ were determined using a random neural network, ensuring diverse functional representations

[4] `https://networks.skewed.de/net/messal_shale`

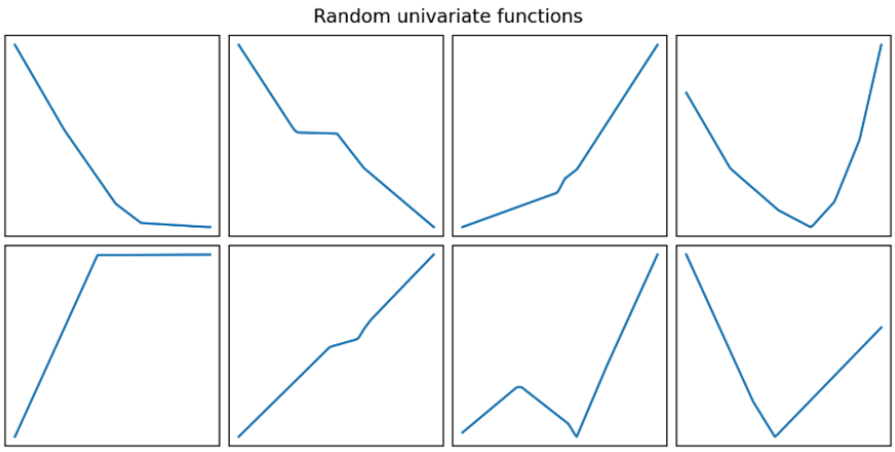


Figure 11: Examples of random univariate functions generated using neural networks with random weights (before adding noise).

C.4.3 Gaussian Processes

Gaussian processes with a Matérn 3/2 kernel were employed to define smooth, continuous functions between variables, providing yet another class of functional relationships.

C.4.4 Post-processing

To maintain consistent variance and prevent trivial identification of the graph's topological order Reisach et al. [2023, 2021], all outputs were uniformized to the range $[-1, +1]$. This step ensured that the functional relationships did not artificially reveal the causal structure through variance patterns or magnitude differences.

C.5 Dataset Generation Process

The final dataset creation involved systematically combining the graph structures and random functions described above:

1. **Graph Selection**: For each dataset, a graph structure was selected from the pool of synthetic and real-world derived graphs.
2. **Function Assignment**: For each node in the graph, a random function was assigned to compute its value based on its parent nodes. The function was selected from one of the three types described (neural networks, Bayesian models, or Gaussian processes).
3. **Noise Introduction**: Gaussian noise was incorporated into the functional outputs, with standard deviations selected from the range $[0, 1]$, simulating the stochasticity present in real-world data.
4. **Data Generation**: Values for each node were computed in topological order according to the assigned functions and parent values, creating a dataset of approximately 1,000 observations.
5. **Output Normalization**: The generated values were uniformized to the range $[-1, +1]$ to maintain consistent variance across variables.

This process was repeated to create the full collection of 47,000 datasets, split between training and test sets. The training datasets included the ground truth causal graphs that generated them, while test datasets withheld this information for evaluation purposes.

C.6 Conclusion

The dataset generation methodology described here ensured a comprehensive suite of causal discovery scenarios, incorporating diverse graph structures, functional relationships, and noise levels. By combining synthetic models with real-world graph structures and employing various function types, the datasets effectively simulated the complexity and challenges present in real-world causal discovery tasks. This robust approach provided an ideal testing ground for evaluating and advancing causal discovery algorithms through the ADIA Lab Causal Discovery Challenge.

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